

Lesson 1 Practical Application Guide

□ How to Run the Project

Step 1: Start Development Server

```
cd D:\learnlaravel2025\lessons\lesson-01\practice-app
php artisan serve
```

Server will run on: <http://localhost:8000>

□ Available Routes

1. Home Page

- **URL:** <http://localhost:8000/>
- **Description:** Default Laravel welcome page

2. Hello Route (Exercise 3)

- **URL:** <http://localhost:8000/hello>
- **Description:** Simple route displaying a greeting message
- **Code:**

```
Route::get('/hello', function () {  
    return '<h1>Hello, I\'m learning Laravel!</h1>';  
});
```

3. About Page (Challenge)

- **URL:** <http://localhost:8000/about>
- **Description:** Full HTML page introducing the developer
- **Features:**
 - Complete design with HTML & CSS
 - Arabic content with RTL support
 - Link back to home page

4. Contact Page (Challenge)

- **URL:** <http://localhost:8000/contact>
- **Description:** Contact information page
- **Content:**
 - Email address
 - Phone number
 - Website URL

5. My Page (Exercise 5 - Blade View)

- **URL:** <http://localhost:8000/mypage>
- **Description:** First complete Blade page
- **Features:**
 - Uses Blade Template Engine

- Displays current date/time using `{{ date() }}`
 - Professional design with gradient background
 - Dynamic content
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▣ Completed Exercises

Exercise 1: Project Exploration ▣

- [x] Identify entry point: `public/index.php`
- [x] Identify routes location: `routes/web.php`
- [x] Identify views location: `resources/views/`
- [x] Review `.env` file

Exercise 2: Start Development Server ▣

```
php artisan serve
```

Exercise 3: Create First Route ▣

- Created `/hello` route

Exercise 3 (Challenge): Additional Routes ▣

- Created `/about` route
- Created `/contact` route

Exercise 4: Explore Artisan Commands

```
# Display all routes
php artisan route:list

# Clear cache
php artisan cache:clear

# Display all commands
php artisan list

# Display Laravel version
php artisan --version
```

Exercise 5: Create Blade View

- Created `resources/views/mypage.blade.php`
- Added `/mypage` route

What We Learned

1. Routes

- How to define simple GET routes
- Return HTML directly from route
- Return Views from route

2. Views

- Create Blade files with `.blade.php` extension
- Use Blade syntax: `{{ }}`

- Pass and display data

3. Laravel Structure

- Understanding `routes/` folder
- Understanding `resources/views/` folder
- Understanding request lifecycle

4. Artisan Commands

- `php artisan serve` - Start server
 - `php artisan route:list` - Display routes
 - `php artisan --version` - Display version
-

□ Useful Commands

```
# Start server
php artisan serve

# Display all routes
php artisan route:list

# Display Laravel info
php artisan about

# Clear all cache
php artisan optimize:clear

# Display help
php artisan help

# Generate application key
php artisan key:generate
```

□ Testing Routes

Visit each route to verify it works:

1. □ <http://localhost:8000/> - Home page
 2. □ <http://localhost:8000/hello> - Hello message
 3. □ <http://localhost:8000/about> - About page
 4. □ <http://localhost:8000/contact> - Contact page
 5. □ <http://localhost:8000/mypage> - Blade page
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□ Next Steps

After successfully completing this lesson, you are now ready for:

1. **Lesson 2:** Routing Basics

- Route Parameters
- Named Routes
- Route Groups
- Route Methods (POST, PUT, DELETE)

2. **Lesson 3:** Controllers

- Creating Controllers
- Resource Controllers
- MVC Pattern

□ Tips

1. **Always use** `php artisan route:list` to view all available routes
2. **Save changes** before refreshing browser
3. **Use Ctrl+C** to stop development server
4. **Read error messages carefully** - Laravel provides clear and helpful error messages

☐ Summary of Achievements

☐ Created complete Laravel project ☐ Created 4 custom routes ☐ Created first Blade view ☐ Understood basic Laravel structure ☐ Used Artisan commands

Congratulations! You now understand Laravel basics! ☐

☐ Help

If you encounter any issues:

1. Check that development server is running (`php artisan serve`)
 2. Verify the route is correct
 3. Check `routes/web.php` file
 4. Review error messages in Terminal
 5. Review theoretical lesson in `README.md`
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☐ Project Structure

```
practice-app/
├── app/                # Core application code
│   ├── Http/
│   │   ├── Controllers/ # Controller classes
│   │   └── Middleware/  # Middleware
│   └── Models/          # Eloquent models
├── routes/
│   └── web.php          # All web routes defined here
├── resources/
│   └── views/
│       ├── welcome.blade.php # Default welcome page
│       └── mypage.blade.php  # Custom Blade view
├── public/             # Entry point & public files
│   └── index.php        # Application entry point
├── database/           # Migrations, seeders, factories
├── .env                # Environment configuration
└── artisan              # Artisan CLI tool
```

📦 Features

📦 **5 Custom Routes** - Home, Hello, About, Contact, MyPage 📦 **Professional Blade View** - With dynamic content 📦 **Full Arabic Support** - RTL and Arabic text 📦 **Dynamic Content** - Using Blade syntax 📦 **Documented Code** - Clear comments and structure

📦 Advanced Artisan Commands

```
# View detailed route information
php artisan route:list -v

# Clear specific caches
php artisan cache:clear      # Clear application cache
php artisan config:clear     # Clear configuration cache
php artisan route:clear      # Clear route cache
php artisan view:clear       # Clear compiled views

# Generate key for new installation
php artisan key:generate

# Run in production mode
php artisan serve --host=0.0.0.0 --port=8080

# Display application information
php artisan about
```

□ Understanding the Code

Route Definition

```
Route::get('/path', function () {
    return 'Response';
});
```

- `Route::get()` - Defines a GET route
- `/path` - The URL path
- `function()` - Closure (anonymous function)
- `return` - What to send back to browser

Blade Syntax

```
{{ $variable }}           // Echo variable
{{ date('Y-m-d') }}       // Echo function result
@if($condition)           // Control structure
```

□ Key Concepts Learned

1. MVC Pattern

- **Model:** Data and database logic (not used yet)
- **View:** Presentation layer (Blade templates)
- **Controller:** Business logic (will learn in Lesson 3)

2. Request Lifecycle

```
Browser → public/index.php → Bootstrap → Route → Middleware → Response
```

3. Blade Template Engine

- `.blade.php` extension
 - `{{ }}` for echoing
 - Directives like `@if`, `@foreach`, etc.
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🔧 Debugging Tips

Check if server is running

```
php artisan serve
```

View all routes

```
php artisan route:list
```

Check Laravel version

```
php artisan --version
```

Clear all caches

```
php artisan optimize:clear
```

📌 Best Practices

1. **Organize Routes:** Group related routes together
2. **Use Comments:** Document your routes
3. **Consistent Naming:** Use clear, descriptive names
4. **Keep It Simple:** Start with simple routes, add complexity later

▢ Additional Resources

- [Official Laravel Documentation](#)
 - [Laravel News](#)
 - [Laracasts](#) - Video tutorials
 - [Laravel Daily](#)
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▢ Practice Exercises

Try these on your own:

1. Create a `/services` route showing your services
 2. Create a `/portfolio` route with your projects
 3. Create a `/testimonials` route with client reviews
 4. Add navigation links between all pages
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▢ Checklist Before Next Lesson

- ☐ Can start Laravel server
- ☐ Understand how to define routes

- [] Can create Blade views
 - [] Know basic Artisan commands
 - [] Understand project structure
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Happy Learning! ☐

Ready for Lesson 2: Routing Basics!